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AP Biology Unit 3: Cellular Energetics

Enzyme Inhibition Rapid-Comparison Sheet + Practice Check

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Why Enzyme Inhibitors Behave Differently on AP Graphs

Two inhibitor types show up constantly on unit tests and AP free-response questions, and it is easy to mix them up if you only memorize "blocks the enzyme."

Feature	Competitive Inhibitor	Allosteric (Noncompetitive) Inhibitor
Where it binds	Directly inside the **active site** (competes with substrate)	At a separate **allosteric site** away from the active site
Mechanism	Blocks substrate entry by occupying the same binding pocket	Changes the enzyme's 3D conformation, distorting the active site geometry
Effect of adding MORE substrate	**Overcomes inhibition:** high [S] outcompetes the inhibitor and restores normal V_{max}	**Does NOT overcome inhibition:** active site stays distorted; V_{max} remains permanently lower
Graph signature	Curve shifts right (higher apparent K_m), but reaches the **same maximum rate**	Initial slope and plateau are both lower; **maximum rate stays capped**

Pathway-Level Rule (Feedback Inhibition): When the end-product of a metabolic pathway accumulates, it binds an early enzyme at an **allosteric site** to throttle the entire pathway automatically so the cell does not waste ATP overproducing a molecule it already has in excess.

Quick AP Exam Check

Question: An enzyme's maximum reaction rate drops by 45% when Compound X is added. Increasing substrate concentration 10-fold does not restore the maximum rate. Is Compound X a competitive or allosteric inhibitor, and why?

Answer & Scoring Logic: *Allosteric (noncompetitive) inhibitor.* Because Compound X binds outside the active site and alters the enzyme's 3D conformation, the active site itself is nonfunctional regardless of how crowded the solution is with substrate.

What's Inside the Complete 10-Page Unit 3 Study Guide

- **Part I — Bioenergetics & ATP:** Free energy (ΔG), activation energy, ATP coupling, and redox reactions
- **Part II — Enzyme Kinetics:** Temperature, pH, salinity, competitive vs. allosteric inhibition, and experimental controls

- **Part III — Photosynthesis:** Light reactions (linear vs. cyclic electron flow), thylakoid proton gradient, and the Calvin cycle
- **Part IV — Cellular Respiration:** Glycolysis, pyruvate oxidation, citric acid cycle, ETC & chemiosmosis, and per-glucose ATP table
- **Part V — Fermentation & C3/C4/CAM Comparisons:** Full side-by-side comparison tables + 10 common AP traps
- **Part VI–VIII — Exam Practice & Cram Sheet:** Practice MCQs, scored FRQs with full rubrics, glossary, and 1-page night-before cram sheet

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Full Unit 3 & Unit 4 Study Guides available at: **https://jerryhsieh991-lang.github.io/ap-bio-study-guides/**